Unit 4 Progress Check: MCQ

A student combines a solution of $NaCl(aq)$ with a solution of $AgNO_3(aq)$, and a precipitate forms.

1. Which of the following is the balanced net ionic equation for the formation of the precipitate?

(A) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

(B) $Na^+(aq) + NO_3{}^-(aq) \rightarrow NaNO_3(s)$

(C) $NaCl(aq) + AgNO_3(aq) \rightarrow NaNO_3(s) + AgCl(aq)$

(D) $NaCl(aq) + AgNO_3(aq) \rightarrow NaNO_3(aq) + AgCl(s)$

2. Which of the following is evidence that ionic bonds formed during the precipitation?

(A) The resulting solution is colorless.

(B) The resulting solution conducts electricity.

(C) The precipitate has a high melting point.

(D) The temperature of the solution did not change significantly during the precipitation.

3. Assume that 50.0 mL of $1.0\ M$ $NaCl(aq)$ and 50.0 mL of $1.0\ M$ $AgNO_3(aq)$ were combined. According to the balanced equation, if 50.0 mL of $2.0\ M$ $NaCl(aq)$ and 50.0 mL of $1.0\ M$ $AgNO_3(aq)$ were combined, the amount of precipitate formed would

(A) double, because all of the coefficients are 1

(B) double, because the amount of one of the reactants was doubled

(C) not change, because all of the coefficients are 1

(D) not change, because the amount of $AgNO_3(aq)$ did not change

4. $Na_2CO_3(aq) + 2\ HCl(aq) \rightarrow 2\ NaCl(aq) + H_2O(l) + CO_2(g)$

A student combined two colorless aqueous solutions. One of the solutions contained Na_2CO_3 as the solute and the other contained HCl. The chemical reaction that took place is represented by the equation above. What experimental result would be evidence that a chemical reaction took place when the solutions were combined?

(A) Bubbles formed when the two solutions were combined.

(B) The total volume of the mixture is close to the sum of the initial volumes.

(C) The resulting solution is colorless.

(D) The resulting solution conducts electricity.

5. A chemistry teacher carried out several demonstrations, and students recorded their observations. For one of the demonstrations, a student concluded that a physical change took place, but not a chemical change. Which of the following observations could the student have made of the results of the demonstration?

(A) Two colorless solutions were combined, and the resulting solution was pink.

(B) When a solid was added to a liquid, sparks were produced.

(C) One piece of solid substance was changed into small pieces.

(D) When two solutions were combined, a precipitate formed.

6. A student was given two clear liquids; a colorless liquid and a dark-blue liquid. The student was asked to combine the liquids in a beaker and record observations. Which of the following results, if true, would provide the best evidence that a chemical change took place when the liquids were combined?

(A) The resulting mixture was cloudy.

(B) The total volume of the mixture was equal to the sum of the initial volumes.

(C) The resulting liquid was light blue.

(D) The liquids formed two separate layers in the beaker.

7. Which of the following is the correct net ionic equation of the neutralization reaction between hydrofluoric acid and sodium hydroxide in aqueous solution?

(A) $HF(aq) + NaOH(aq) \rightarrow H_2O(l) + NaF(aq)$

(B) $HF(aq) + NaOH(aq) \rightarrow NaH(aq) + FOH(aq)$

(C) $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

(D) $HF(aq) + OH^-(aq) \rightarrow H_2O(l) + F^-(aq)$

8. Which of the following is the correct net ionic equation of the reaction that occurs when aqueous solutions of ammonia and hydrochloric acid are combined?

(A) $NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)$

(B) $NH_3(aq) + H^+(aq) \rightarrow NH_4{}^+(aq)$

(C) $NH_3(aq) + HCl(aq) \rightarrow NH_4{}^+(aq) + Cl^-(aq)$

(D) $NH_3(aq) + H^+(aq) + Cl^-(aq) \rightarrow NH_4{}^+(aq) + Cl^-(aq)$

9. A beaker was half filled with freshly distilled H_2O and placed on a hot plate. As the temperature of the water reached $100\,^\circ C$, vigorous bubbling was observed in the beaker. The gaseous contents of the bubbles were analyzed. The presence of which of the following substances would support the claim that the observed phenomenon was a physical change?

(A) $H_2(g)$

(B) $O_2(g)$

(C) $CO_2(g)$

(D) $H_2O(g)$

10.

Mass of NaI crystals	2.0 g
Color of NaI crystals	White
Color of solid recovered	Yellow
Color of filtered solution	Colorless
Mass of dry solid recovered	3.0 g

When students added 2.0 g of NaI crystals to $100.$ mL of $Pb(NO_3)_2(aq)$, a yellow precipitate formed. After the solution was filtered, the yellow solid was dried and weighed. Data from the experiment are shown in the table above. Which of the following claims is best supported by the observations?

(A) A physical change occurred when a new yellow compound was formed.

(B) A physical change occurred when the color of the NaI solid added changed to yellow when mixed with water.

(C) A chemical change occurred when a yellow, insoluble compound with a larger mass than the original NaI formed.

(D) A chemical change occurred when covalent bonds between the yellow solid and water were broken during drying.

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11. $HC_2H_3O_2(aq) + OH^-(aq) \rightarrow C_2H_3O_2^-(aq) + H_2O(l)$

A student carried out a titration using $HC_2H_3O_2(aq)$ and $NaOH(aq)$. The net ionic equation for the neutralization reaction that occurs during the titration is represented above. The $NaOH(aq)$ was added from a buret to the $HC_2H_3O_2(aq)$ in a flask. The equivalence point was reached when a total of $20.0\ mL$ of $NaOH(aq)$ had been added to the flask. How does the amount of $HC_2H_3O_2(aq)$ in the flask after the addition of $5.0\ mL$ of $NaOH(aq)$ compare to the amount of $HC_2H_3O_2(aq)$ in the flask after the addition of $1.0\ mL$ of $NaOH(aq)$, and what is the reason for this result?

(A) It is less because more $HC_2H_3O_2(aq)$ reacted with the base.

(B) It is the same because the half-equivalence point has not been reached.

(C) It is the same because all of the coefficients in the neutralization equation are 1.

(D) It is greater because $HC_2H_3O_2(aq)$ is a proton donor.

12. $K_2SO_3(aq) + 2\ HNO_3(aq) \rightarrow 2\ KNO_3(aq) + SO_2(g) + H_2O(l)$

According to the balanced chemical equation above, when $100.0\ mL$ of $0.100\ M\ K_2SO_3(aq)$ is mixed with $100.0\ mL$ of $0.200\ M\ HNO_3(aq)$ at $30^\circ C$ and $1\ atm$, the volume of SO_2 gas produced is $0.24\ L$. If it is assumed that the reaction goes to completion, which of the the following changes would double the volume of SO_2 produced at the same temperature and pressure? (For each change, assume that the other solutions and volumes remain the same.)

(A) Using $200.0\ mL$ of the $0.100\ M\ K_2SO_3(aq)$, because it then becomes the reactant in excess

(B) Using $200.0\ mL$ of the $0.200\ M\ HNO_3(aq)$, because the volume of SO_2 produced is inversely proportional to the number of moles at constant temperature and pressure

(C) Using $200.0\ mL$ of $0.100\ M\ K_2SO_3(aq)$ and $200.0\ mL$ of $0.200\ M\ HNO_3(aq)$, because these are the required stoichiometric amounts

(D) Using $400.0\ mL$ of $0.100\ M\ K_2SO_3(aq)$ and $200.0\ mL$ of $0.200\ M\ HNO_3(aq)$, because this provides the same number of moles of each reactant

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13. $Zn(s) + 2\,HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

When the reaction represented above proceeds, heat is produced. Which of the following best describes the reaction?

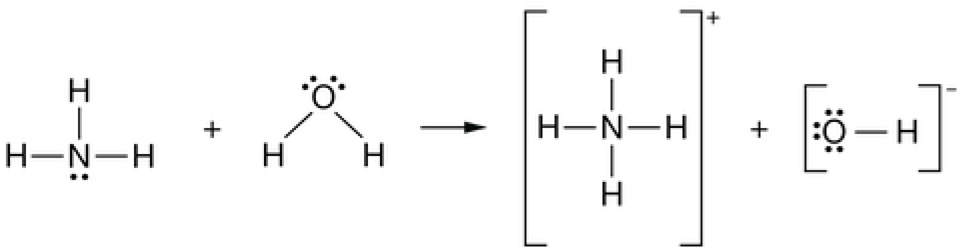
(A) It is a combustion reaction because heat is produced by the reaction.

(B) It is a double replacement reaction because 2 Cl atoms are added to Zn.

(C) It is an acid-base reaction because HCl is an acid that is capable of exchanging H^+.

(D) It is an oxidation-reduction reaction because zinc is oxidized and hydrogen is reduced.

14.



Which of the following best describes the process represented above that takes place when NH_3 is added to water?

(A) It is a single replacement reaction in which an electron pair on N is replaced with an H atom.

(B) It is an acid-base reaction in which a proton is exchanged from H_2O to NH_3.

(C) It is a precipitation reaction in which NH_4OH, an insoluble solid, is produced.

(D) It is an oxidation-reduction reaction in which the oxidation number of N changes from -3 to -4.

15. $2\,Mg(s) + SiCl_4(l) \rightarrow 2\,MgCl_2(s) + Si(s)$

Which of the following statements about the reaction represented above is correct?

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(A) It is an oxidation-reduction reaction, and Mg is oxidized.

(B) It is an oxidation-reduction reaction, and electrons are transferred from $SiCl_4$ to Mg

(C) It is an oxidation-reduction reaction, and the oxidation number of Cl changes from $+4$ to $+2$.

(D) It is not an oxidation-reduction reaction because none of the oxidation numbers change.

16. $C_6H_5COOH(aq) + NaOH(aq) \rightarrow C_6H_5COONa(aq) + H_2O(l)$

Which of the following identifies a conjugate acid-base pair in the reaction represented above?

(A)

Acid	Conjugate base
C_6H_5COOH	OH^-

(B)

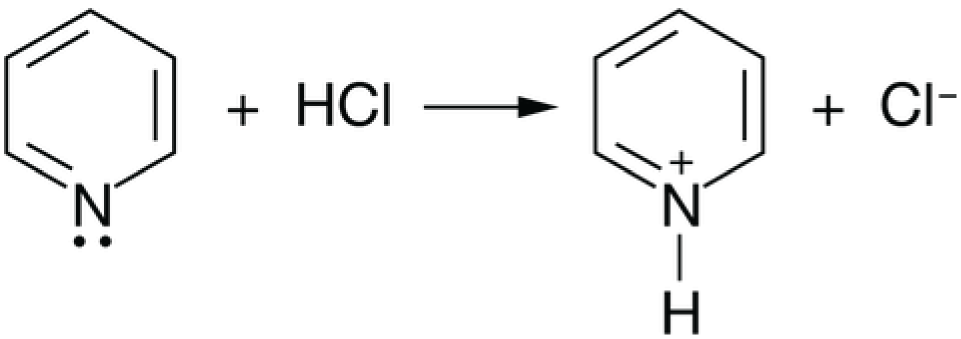
Acid	Conjugate base
C_6H_5COOH	$C_6H_5COO^-$

(C)

Acid	Conjugate base
OH^-	H_2O

(D)

Acid	Conjugate base
OH^-	$C_6H_5COO^-$

17. 

In the reaction between $C_5H_5N(aq)$ and $HCl(aq)$ represented above, C_5H_5N acts as

(A) a Brønsted-Lowry base

(B) a Brønsted-Lowry acid

(C) the conjugate base of HCl

(D) the conjugate acid of $[C_5H_5NH]^+$

18. Based on the Brønsted-Lowry theory of acids and bases, which of the following species can act as both a conjugate acid and a conjugate base?

(A) HS^-

(B) CH_3COO^-

(C) H_3O^+

(D) $NH_4{}^+$

19. $5\ Fe^{2+}(aq) + MnO_4{}^-(aq) + 8\ H^+(aq) \rightarrow 5\ Fe^{3+}(aq) + Mn^{2+}(aq) + 4\ H_2O(l)$

Which of the following represents the oxidation half-reaction based on the balanced ionic equation shown above?

(A) $Fe^{2+}(aq) + e^- \rightarrow Fe^{3+}(aq)$

(B) $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

(C) $MnO_4^-(aq) \rightarrow Mn^{2+}(aq) + 5\ e^-$

(D) $MnO_4^-(aq) + 5\ e^- \rightarrow Mn^{2+}(aq)$

20. $Al(s) \rightarrow Al^{3+}(aq) + 3\ e^-$
$Sn^{2+}(aq) + 2\ e^- \rightarrow Sn(s)$

Based on the half-reactions represented above, which of the following is the balanced ionic equation for the oxidation-reduction reaction between $Al(s)$ and $Sn^{2+}(aq)$?

(A) $Al(s) + Sn^{2+}(aq) \rightarrow Al^{3+}(aq) + Sn(s) + e^-$

(B) $3\ Al(s) + 2\ Sn^{2+}(aq) \rightarrow 3\ Al^{3+}(aq) + 2\ Sn(s)$

(C) $2\ Al(s) + 3\ Sn^{2+}(aq) \rightarrow 2\ Al^{3+}(aq) + 3\ Sn(s)$

(D) $Al(s) + Sn^{2+}(aq) + 2\ e^- \rightarrow Al^{3+}(aq) + Sn(s) + 3\ e^-$

21. $Zn(s) + CuSO_4(aq) \rightarrow Cu(s) + ZnSO_4(aq)$

When a zinc plate is placed in an aqueous solution of copper sulfate, elemental copper forms, as represented by the equation above. Which of the following represents the reduction half-reaction of the reaction?

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(A) $Cu^{2+}(aq) + 2\,e^- \rightarrow Cu(s)$

(B) $Cu(s) \rightarrow Cu^{2+}(aq) + 2\,e^-$

(C) $Zn(s) + 2\,e^- \rightarrow Zn^{2-}(aq)$

(D) $Zn(s) \rightarrow Zn^{2+}(aq) + 2\,e^-$